Faculty of Engineering & Information Technology
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31338 Network Servers
Lab 5a: Managing users and groups

Aims
1. To add, modify and delete users and groups using command-line tools
2. To understand the actions that the command-line tools are performing and be able to add users manually

Task 1: Creating users with command-line tools

Using the command-line tools (`useradd`, etc), create a username *peter* with the default uid and gid. Peter's full name is "Peter Griffin" and his preferred login shell is `zsh` (the Z shell). With the Fedora/Redhat/Centos version of useradd, you should find that a default home directory is created for Peter, and that the skeleton files are copied, without further options than those for the shell and the comment field. Set a password for Peter. Note that you can set a password for new users by an option of the useradd command. But in practice this is seldom or never done. Read the man pages for useradd to try to ascertain why the password option is inconvenient. Instead, use the passwd command as root to set initial passwords for all new accounts. Make appropriate notes in your learning journal.

You've decided that when you create home directories for all new users, you want to supply a `README` file by default, in the user's home directory. Modify the skeleton directory (`/etc/skel`) so that it contains a `README` file (the contents are unimportant).

Now create a second user with username *stewie* with the default uid and gid. Stewie's full name is "Stewie Griffin", and his preferred login shell is `bash`. Make sure that the default home directory is created for Stewie and that the new `README` file is copied into it when the account is created. Set a password for Stewie.

Create a third user with username *brian* with uid 200 and make Brian's default group to be the group called "`users`". Do not let Linux create a "`brian`" group when the account is created (which is the default behaviour). Brian's full name is "Brian Griffin", and his preferred login shell is `bash`. Set a password for Brian.

Test that you can log in as each of these users.
Some ways to try:
1. GUI: System → Logout (or Switch User) (What has changed about the login screen?)
2. Command line: ssh brian@localhost

Check the group memberships with the "id" command.

Task 2: Configuring groups and assigning users to groups

Create a group called *family*, using the `groupadd` command. Make Stewie and Brian members of this group (it should not be their primary group though), using the `usermod -G` command (note: capital G). Log in as Stewie or Brian and verify their group membership by running the "`id`" command. Create a new file (for example, by using the touch command). Who is the group owner? Run the ps command, and note down the PID of your shell. You will use this information later. While logged in as Stewie or Brian, change their current group using the "`newgrp`" command, and verify the change with "`id`". Now create another file and check its group ownership. What do you observe? There are many subtleties to be noted in this task. For example, what is the mechanism by which "newgrp" works? Try to ascertain this by reading the man pages. Use the ps command to see how many shells you have running now. What would be the most sensible way to reverse the effects of newgrp? In other words, how do you get back to the previous situation? In your journal, describe the difference (if any) between a user's default (or primary) group, and other group memberships.

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Task 3: Modifying user account settings

Modify Peter's account so that it will expire in 5 days. You should try logging in as Peter next week.
(hint: `chage -E` or `usermod -e` – they do the same thing)

Check Peter's password aging parameters with: `chage -l peter`

Modify Stewie's account so that he is required to change his password at least every 5 days. You should also try logging in as Stewie next week to test his password aging (hint: `chage -M`). Check using the `chage` command.

Lock Brian's account so he cannot log in. Try doing it three ways:

- Lock Brian's account with the usermod command (hint: `usermod -L`)
- by "locking" Brian's password (putting a `*` in it, or putting a `!!` in front of the password). Then unlock it this way
  Note: the correct way to directly edit the shadow file is with: `vipw -s`

> Further note: if you really cannot use vi you can make vipw use a different editor. First run:
> `export EDITOR=gedit`
> and then run vipw afterwards (or any other plain text editor works too, e.g. `nano`). However, having some basic capability with vi is useful for system administrators as a GUI is not always available, e.g. remote server administration)

- by changing Brian's shell to a "`/sbin/nologin`" shell, and change the default message so that it says "Please contact the system administrator to have your account re-enabled". (Hint: `man nologin` and use `usermod` to set Brian's shell to `/sbin/nologin`).

Task 4: Creating a user without using command-line tools

Finally, create a fourth user with username *lois*. However this time, do not use the `useradd` command – perform each of the steps manually. Of course in practice you would never do this – it is just so you have confidence in knowing what all the steps are and exactly what's going on behind the scenes when you use the useradd command.

Remember to use `vipw` when editing the passwd file and `vigr` to edit the group file. Lois's full name is "Lois Griffin", and she should be allocated the next uid and gid in sequence, as Linux would have done. Lois prefers the `bash` shell. Don't forget to set her password (using the `passwd` command), create a home directory (with skeleton files), and `chown` her home directory and skeleton files.

Test that you have created the account properly by logging in as Lois. Make the following checks:

- Does lois have a home directory in the right location?
- Who is the user owner and group owner of lois' home directory?
- What files are present in lois' home directory?
- How did you ensure that you copied the dot files to lois' home directory?
- What are the ownerships of these files?

Task 5: Using the GUI

You can use the graphical user/group administration GUI to do all of the above steps.
Check and confirm the configuration of lois using the GUI via Settings → Details → Users.
What details can't you see or set via the GUI?

Make notes in your journal.

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Task 6: Using Windows Server GUI to add users and groups

We will not use Active Directory to manage our Windows Server, and for the moment will simply modify local groups.

Start with Server Manager → Local Server. From the Tools menu, choose "Computer Management" and on the left you should see "Local Users and Groups". Expand this so you see separate menus for Users and Groups.

Right click on "Users" and choose "New User". Add the above users (peter, stewie, brian, lois). Then do the same thing with "Groups" and add their respective group (family) and assign them as members of the group.

When adding users to a group, you have to search for the appropriate "object" to add as a member. Use the Add button to do this which produces a "Select Users" popup.

You then need to type the object name (e.g. brian) into the entry field , then press **Check Names** to confirm it.
ALTERNATIVELY: Click **Advanced** and you should get a detailed search panel. Choosing **Find Now** will show all the valid objects you can add, so select brian from this list.

Notice that the user name is of the format: *workstation\user.*
In a domain environment, the *workstation* prefix would be replaced by the Domain name.

Task 7: Using Windows Server command line to add users and groups

As with many things on Windows Server there are PowerShell modules and normal Windows commands that can achieve similar results.

Some PowerShell commands to try (you may to run PowerShell as Administrator):

- `New-LocalUser -Name "joe" -FullName "Joe Swanson"`
- `Get-LocalUser`
- `Disable-LocalUser joe`
- `Enable-LocalUser joe`

See also:

- **New-LocalUser** (and browse other command documentation from the menu)

Some Windows Command Prompt commands to try:

- `net user bonnie /add /fullname:"Bonnie Swanson"`
- `net user`
- `net user bonnie`
- `wmic useraccount where "name='bonnie' "`

See also:

- **Administrative tasks from the command line**
- **Net user command**

Make notes in your journal about these commands and options.

Remember that although you can achieve everything using the Windows GUI, the benefit of being familiar with some of the command-line tools is that they are SCRIPTABLE. As a system administrator you can create scripts that could for example create hundreds of user accounts at once from data in an Excel spreadsheet.